

Reality :- Null Hypothesis (H_0) is True or Null hypothesis is False

Decision :- Null hypothesis (H_0) is True or Null hypothesis is False.

Outcome ① $\Rightarrow$ we reject the null hypothesis where in reality it is False $\Rightarrow$ Good

Outcome ② $\Rightarrow$ we reject null hypothesis where in reality it is True $\Rightarrow$ Type 1 Error

Outcome ③ $\Rightarrow$ we retain null hypothesis where in reality it is false $\Rightarrow$ Type 2 Error

$\downarrow$
we failed to reject null hypothesis where in reality it is false.

Outcome ④ $\Rightarrow$ we retain null hypothesis where in reality it is true (Good)

we fail to reject null hypothesis where reality it is true (Good)